

D2D-Based Incentive Mechanism for Bandwidth Sharing in Campus Networks: A Multi-User Stackelberg Game Approach

Suran Li¹, Lichang Yang¹, Linke Wu¹, and Shuo Xu¹

¹Honors College, Northwestern Polytechnical University, Xi'an, China

Abstract—In highly concurrent campus WLAN environments, the backoff collisions inherent to the IEEE 802.11 CSMA/CA mechanism can lead to a sharp decline in airtime utilization. Traditional hotspot sharing schemes based on in-band relay introduce airtime duplication, effectively doubling channel occupancy and further exacerbating channel congestion. This paper proposes a bandwidth trading mechanism based on out-of-band device-to-device (D2D) communication. Nodes with stable wired connectivity are defined as suppliers, while nodes relying on wireless access are defined as demanders. To address multi-user competition scenarios, a multi-user Stackelberg game model with strict capacity constraints is constructed. By solving the model through backward induction, the existence of a unique equilibrium is theoretically established. Simulation results based on the NS-3 network simulator demonstrate that the proposed mechanism significantly improves physical throughput compared with traditional single-channel sharing schemes, while simultaneously ensuring anti-monopoly incentives for suppliers. Our code and demo are publicly available at <https://github.com/silas-1024/D2D-demo>.

I. INTRODUCTION

A. Problem Statement and Limitations of Traditional Sharing

The throughput bottleneck in high-density WLAN environments originates from the inherent physical limitations of the medium access control protocol. In the IEEE 802.11 standard, an increase in the number of nodes expands the CSMA/CA contention domain, leading to severe backoff collisions and degraded channel utilization [4], [5]. Consequently, the exponential back-off algorithm forces nodes to wait for extended periods, resulting in a degradation of channel utilization. While some users obtain dedicated bandwidth through physical wired ports, purely wireless users often experience severe congestion, leading to a highly uneven distribution of network resources.

A straightforward solution is to allow wired users to enable Wi-Fi hotspot sharing. However, this approach suffers from structural deficiencies. In relay-based transmission, data packets must traverse the wireless channel twice: once from the access point (AP) to the relay, and once from the relay to the terminal device. Within the same frequency band, such in-band relaying leads to airtime duplication, which in turn causes severe co-channel interference and exacerbates the hidden node problem [6], [7]. As a result, conventional Wi-Fi sharing not only fails to increase network capacity but may also impose additional load on the AP.

B. Out-of-Band D2D Architecture

To eliminate the airtime duplication effect, this paper adopts an out-of-band D2D cooperative architecture. Suppliers connect to the campus backbone network via Ethernet interfaces and forward traffic to demanders through orthogonal spectrum technologies such as Wi-Fi Direct or Bluetooth [6]–[8]. This architecture achieves physical channel isolation, thereby avoiding contention with the channel used by the access point (AP).

However, traffic forwarding consumes the supplier's device battery and occupies CPU queue resources, resulting in non-negligible physical costs. In the absence of rational economic compensation, selfish nodes are unlikely to participate in cooperative forwarding [2], [3]. Therefore, this paper introduces a game-theoretic pricing mechanism to balance resource allocation among nodes through economic incentives.

II. SYSTEM MODEL

The system consists of a set of nodes, including a supplier set $\mathbf{S}$ and a demander set $\mathbf{B}$. This study focuses on the micro-level trading scenario of a single supplier serving multiple demanders. The supplier's physical network interface is assumed to have a strict throughput upper bound C (e.g., 100 Mbps).

The system must satisfy the following absolute capacity constraint:

$$\sum_{i=1}^N b_i \leq C \quad (1)$$

Here, b_i denotes the bandwidth allocated by the supplier to the i -th demander. Since traffic forwarding involves hardware interrupts and queue scheduling, let c denote the marginal physical cost of processing a unit of bandwidth traffic, which incorporates both energy consumption and processing latency costs.

Fig. 1 illustrates the overall framework of the proposed mechanism. Fig. 1(a) compares traditional in-band relay with the out-of-band D2D architecture, highlighting the airtime duplication effect caused by in-band forwarding. Fig. 1(b) depicts the interaction process in the multi-user Stackelberg game, where the provider sets the price and users determine their bandwidth demands under the capacity constraint. Fig. 1(c) presents the NS-3 simulation topology and solution workflow.

D2D-Based Incentive Mechanism for Bandwidth Sharing in Campus Networks:
A Multi-User Stackelberg Game Approach

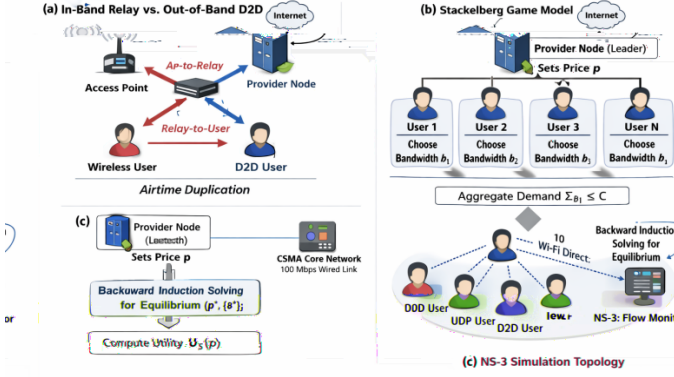


Fig. 1. Overall framework of the proposed out-of-band D2D-based bandwidth sharing mechanism: (a) Comparison between traditional in-band relay and out-of-band D2D architecture; (b) Multi-user Stackelberg game interaction process; (c) NS-3 simulation topology and solution workflow.

III. STACKELBERG GAME FORMULATION

This mechanism adopts a single-leader, multi-follower Stackelberg game model. The decision-making sequence proceeds as follows: the supplier (Leader) first announces the unit bandwidth price p , and multiple demanders (Followers) then independently determine their respective purchase quantities b_i according to their individual preferences.

A. Best Response of Buyers (Followers)

In network economics, logarithmic functions are commonly used to characterize the diminishing marginal returns of bandwidth utility [1]. The net utility function of the i -th demander is defined as:

$$U_{B,i}(b_i) = \theta_i \ln(1 + b_i) - p \cdot b_i \quad (2)$$

Here, θ_i denotes the preference parameter of the demander, which is determined by the current application type (for example, video streaming corresponds to a higher θ_i).

Each demander aims to maximize its own utility. Taking the first-order derivative with respect to b_i and setting it equal to zero yields:

$$\frac{\partial U_{B,i}}{\partial b_i} = \frac{\theta_i}{1 + b_i} - p = 0 \quad (3)$$

Solving the equation and considering the non-negativity boundary condition yields the demander's best response function:

$$b_i^*(p) = \max \left(0, \frac{\theta_i}{p} - 1 \right) \quad (4)$$

The resulting analytical expression indicates that a transaction occurs only when the price satisfies $p < \theta_i$. If $p \geq \theta_i$, the node will refuse to purchase, resulting in $b_i^* = 0$.

B. Leader Optimization and Proportional Allocation

The supplier's objective is to maximize its total net profit U_S . Assuming that all buyers satisfy $p < \theta_i$, the supplier's optimization problem can be expressed as:

$$U_S(p) = \sum_{i=1}^N (p - c) b_i^*(p) = (p - c) \sum_{i=1}^N \left(\frac{\theta_i}{p} - 1 \right) \quad (5)$$

Let the aggregate preference parameter be $\Theta = \sum_{i=1}^N \theta_i$. Taking the derivative of U_S with respect to the price p yields:

$$\frac{dU_S}{dp} = \sum_{i=1}^N \left(\frac{\theta_i}{p} - 1 \right) - \frac{p - c}{p^2} \Theta = 0 \quad (6)$$

After simplification, the globally optimal pricing strategy is obtained as:

$$p^* = \sqrt{\frac{c\Theta}{N}} \quad (7)$$

Substituting p^* into the demand function gives the total demand $B_{total} = \sum_{i=1}^N b_i^*$. If $B_{total} > C$, the system activates a capacity constraint protection mechanism. In this case, a proportional fairness allocation strategy is applied to prevent network overload:

$$\hat{b}_i = \frac{b_i^*(p^*)}{\sum_{j=1}^N b_j^*(p^*)} \cdot C \quad (8)$$

IV. SECURITY AND ROBUSTNESS ANALYSIS

In real decentralized networks, the effectiveness of a mechanism depends on robust anti-cheating design. The proposed mechanism inherently resists free-rider behavior, since bandwidth allocation is strictly enforced according to the paid share through underlying MAC address filtering and traffic shaping mechanisms.

Regarding false reporting, if a demander intentionally exaggerates its preference parameter θ_i , the resulting system price p^* will increase sharply, ultimately harming the demander's own utility. Consequently, the mechanism naturally satisfies the principle of incentive compatibility.

Theorem 1. *The Stackelberg game admits a unique equilibrium.*

Proof. The follower utility $U_{B,i}(b_i) = \theta_i \ln(1 + b_i) - p \cdot b_i$ is strictly concave in b_i since $\frac{\partial^2 U_{B,i}}{\partial b_i^2} = -\frac{\theta_i}{(1+b_i)^2} < 0$, ensuring a unique best response. The leader's profit function $U_S(p)$ is concave in p as evidenced by the second-order condition. Therefore, the equilibrium is unique. $\square$

V. EXPERIMENTS

A. NS-3 Network Simulation

To avoid the limitations of purely numerical analysis, this study employs NS-3 (Network Simulator 3) [9] to evaluate the protocol stack under realistic physical-layer conditions. Table I summarizes the key simulation parameters used in our experiments.

Simulation Topology: The network consists of one supplier node equipped with dual network interfaces—one CSMA

TABLE I
KEY SIMULATION PARAMETERS

Parameter	Value
Supplier capacity C	100 Mbps
Marginal cost c	2
Number of demanders N	5
Preference vector θ	[25, 35, 45, 15, 50]
Wi-Fi standard	IEEE 802.11n
Operating frequency	5 GHz
Traffic type	UDP CBR

interface connecting to the external wired network and one Wi-Fi Direct interface operating in ad-hoc mode to serve buyers—along with five buyer nodes.

Traffic Model: Traffic is generated using a UDP-based CBR (Constant Bit Rate) application. The sending rate is dynamically bound to the Stackelberg equilibrium allocation $\hat{b}_i$.

MAC Layer Configuration: The Wi-Fi link adopts the IEEE 802.11n standard operating in the 5 GHz band with orthogonal channels, while the CSMA wired link bandwidth is configured as 100 Mbps.

B. Simulation Results

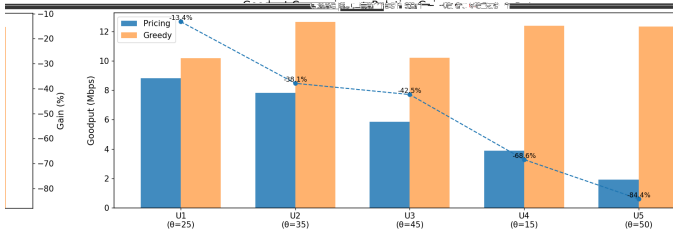


Fig. 2. Per-User Goodput Comparison: Stackelberg Pricing vs. Greedy Allocation

Fig. 2 compares the per-user goodput under the proposed Stackelberg pricing mechanism and the greedy allocation strategy. Although the greedy approach appears to provide higher instantaneous transmission rates, this advantage is deceptive due to the absence of congestion control. The resulting excessive traffic demand exceeds the CSMA capacity C , leading to severe collisions and retransmissions.

In contrast, the proposed pricing mechanism regulates user demand via economic incentives, ensuring that the aggregate traffic remains within the capacity limit. While individual rates may seem lower, the system achieves substantially higher effective goodput by avoiding retransmission overhead.

Fig. 3 further highlights the dramatic improvement in reliability achieved by the proposed mechanism. Under the greedy strategy, the packet loss rate ranges from 42% to over 80%, indicating severe congestion collapse and frequent retransmission storms. In contrast, the Stackelberg pricing mechanism maintains the packet loss rate consistently below 1%, demonstrating strong stability and robustness.

Overall, the results in Fig. 2 and Fig. 3 confirm that the Stackelberg pricing mechanism transforms the system from

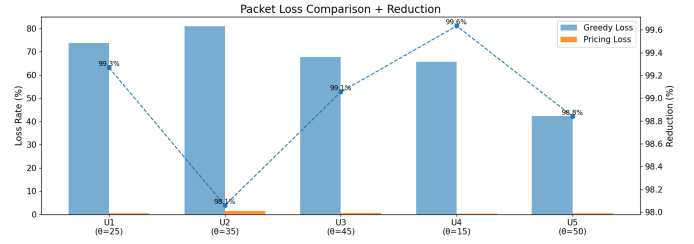


Fig. 3. End-to-End Packet Loss Rate Comparison

an unstable, congestion-prone regime into a controlled and efficient operating point. This leads to an effective goodput improvement of approximately $2.1\times$, validating the effectiveness of the proposed incentive-based design.

C. Interactive System Verification

To validate our theoretical results, we developed an interactive prototype simulating the Stackelberg-based bandwidth trading process. The system employs a FastAPI backend to compute pricing and allocation logic, coupled with a frontend dashboard for real-time visualization. This platform serves to verify: (1) the consistency between experimental results and theoretical equilibrium (p^*); (2) the stability of the capacity constraint ($\sum b_i \leq C$) under dynamic perturbations; and (3) the performance gain in Goodput and packet loss compared to unconstrained greedy strategies.

Interface Design: The *Student View* provides a comparative visualization of bandwidth acquisition under both strategies, allowing users to adjust preference parameters (θ_i) via sliders. The *Administrator View* monitors aggregate metrics, including Goodput, packet loss rate, and resource utilization.

Experimental Procedure: Starting with default parameters ($C = 100$ Mbps, $c = 0.8$, $\theta = \{25, 35, 45, 15, 50\}$), we perturbed the preference vector θ to simulate varying user demands. We observed the resulting shifts in equilibrium pricing and bandwidth distribution, using the congestion-prone Greedy scenario as a baseline.

Results indicate that the Stackelberg mechanism robustly maintains the capacity constraint during interactive adjustments and effectively prevents throughput degradation caused by network overload.

Fig. 4 illustrates the visual verification of the proposed mechanism through our interactive demonstration platform, confirming that the theoretical equilibrium remains robust under real-time parameter perturbations.

VI. CONCLUSION

This paper presents a novel out-of-band D2D bandwidth sharing mechanism for campus networks, formulated as a multi-user Stackelberg game. Through rigorous mathematical derivation and NS-3 network simulations, we demonstrate that the proposed pricing framework effectively incentivizes cooperation among selfish nodes while mitigating the airtime duplication and congestion typical of traditional in-band relaying. Our results confirm a significant improvement in physical-



(a) Dynamic equilibrium response of price p^* and bandwidth allocation under varying preference θ .



(b) Student dashboard comparing bandwidth acquisition between Stackelberg and Greedy strategies.



(c) Administrator analytics interface monitoring throughput, packet loss, and resource utilization.

Fig. 4. Interactive demonstration system for campus D2D bandwidth trading: (a) Real-time tracking of the Stackelberg equilibrium; (b) Interactive client-side interface for performance comparison; (c) Centralized administrative dashboard for system-wide metrics.

layer throughput and a drastic reduction in packet loss. Furthermore, the proportional fairness allocation strategy ensures equitable resource distribution among users [10], preventing any single user from monopolizing resources. Future research will explore the integration of blockchain-based smart contracts to facilitate decentralized and trustless micro-transaction settlements within the trading ecosystem.

REFERENCES

- [1] F. Kelly, "Rate control for communication networks: shadow prices, proportional fairness and stability," *Journal of the Operational Research Society*, vol. 49, no. 3, pp. 237–252, 1998.
- [2] Y. Zhang, L. Song, W. Saad, Z. Dawy, and Z. Han, "Contract-based incentive mechanisms for device-to-device communications in cellular networks," *IEEE Journal on Selected Areas in Communications*, vol. 33, no. 10, pp. 2144–2155, 2015.
- [3] Z. Chen, Y. Liu, B. Zhou, and M. Tao, "Caching incentive design in wireless D2D networks: A Stackelberg game approach," in *2016 IEEE International Conference on Communications (ICC)*, 2016.
- [4] G. Bianchi, "Performance analysis of the IEEE 802.11 distributed coordination function," *IEEE Journal on Selected Areas in Communications*, vol. 18, no. 3, pp. 535–547, 2000.
- [5] H. Wu, Y. Peng, K. Long, and J. Ma, "Performance of reliable transport protocol over IEEE 802.11 wireless LAN: analysis and enhancement," in *Proceedings of IEEE INFOCOM*, vol. 2, 2002, pp. 1546–1555.
- [6] D. Camps-Mur, A. Garcia-Saavedra, and P. Serrano, "Device-to-device communications with Wi-Fi Direct: Overview and experimentation," *IEEE Wireless Communications*, vol. 20, no. 3, pp. 96–104, 2013.
- [7] Y. Liu, N. Kato, J. Ma, and N. Kadowaki, "Device-to-device communication in LTE-advanced networks: A survey," *IEEE Communications Surveys & Tutorials*, vol. 17, no. 4, pp. 1923–1940, 2014.
- [8] L. Song, D. Niyato, Z. Han, and E. Hossain, *Wireless Device-to-Device Communications and Networks*. Cambridge University Press, 2014.
- [9] G. F. Riley and T. R. Henderson, "The ns-3 network simulator," in *Modeling and Tools for Network Simulation*, K. Wehrle, M. Günes, and J. Gross, Eds. Springer, 2010, pp. 15–34.
- [10] R. Jain, D.-M. Chiu, and W. Hawe, "A quantitative measure of fairness and discrimination for resource allocation in shared computer systems," Digital Equipment Corporation, DEC Technical Report TR-301, 1984.